

UI UX PROJECT -

PulseLink 

Emergency Blood Donation & Response Network



Prepared By -

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INTRODUCTION -

- Healthcare emergencies often require quick access to reliable information and support. Finding the required blood group during an emergency can become challenging when people have to depend on personal contacts, social media groups, hospitals, or multiple online sources .
- **Pulselink – blood donation & emergency network** is a UI/UX design concept developed to address this problem. The proposed mobile application aims to provide a simple and centralized platform where users can search for nearby blood donors, filter donors according to blood group, create emergency blood requests, locate blood banks, and manage their donation information .
- The project follows a **human - centered design process**, beginning with problem identification and user research and progressing through defining user needs, ideation, wireframing, high-fidelity interface design, prototyping, and usability testing .
- The primary focus of Pulselink is to create a **simple , accessible , and emergency - focused user experience** that can help users find relevant blood donation support with less confusion and fewer steps .

PURPOSE OF THE PROJECT -

- The purpose of this project is to explore how UI/UX design can be used to improve the experience of people searching for blood donors during emergencies. The design focuses on reducing unnecessary searching, improving access to relevant donor information , and making important emergency actions easy to discover .

Project Scope -

- The project focuses on designing and prototyping the user interface and experience of PulseLink . The main features considered in the prototype include :
 - Emergency blood request
 - Blood donor search
 - Blood group filtering
 - Nearby donor discovery
 - Blood bank access
 - Donor contact options
 - User profile and donation history

CERTIFICATE -



CENTRE FOR
**PROFESSIONAL
ENHANCEMENT**

Certificate No. 491605



Certificate of Merit

This is to certify that Mr./Ms. Anuj Singh S/D/W/o Mr. Rajinder Kumar
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organized by Centre for Professional Enhancement Lovely Professional University
from 14-06-2026 to 26-07-2026 and obtained A Grade.

Date of Issue: 13-08-2026
Place of Issue: Phagwara (India)

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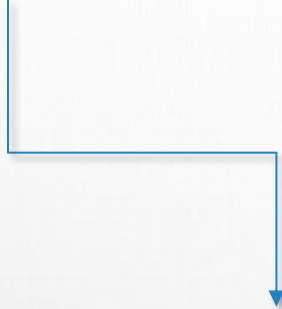
Programme Coordinator
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Head of School
School of Computer Science and Engineering

ACKNOWLEDGEMENT -

- I would like to express my sincere gratitude to my faculty and department for providing me with the opportunity to work on this UI/UX project , “ **PULSELINK – BLOOD DONATION & EMERGENCY NETWORK .** ”
- I am thankful for the guidance and feedback provided throughout the project, which helped me understand and apply the human – centred design process .
- I would also like to thank all the participants who contributed to the user research survey. Their responses and insights helped in understanding the challenges faced by people while searching for blood donors during emergencies .
- Finally , I would like to thank everyone who supported me during the research , design , prototyping , and testing stages of this project .

THE PROBLEM



- Every day, thousands of patients require blood due to accidents, surgeries, childbirth complications, and chronic illnesses. During emergencies, finding the right blood donor often becomes a race against time. Most people still depend on family members, friends, hospitals, or social media platforms to locate donors, which can cause critical delays .

“ A delay of a few minutes can make the difference between life and death. Finding blood should never be the hardest part of saving a life . ”

RESEARCH GOALS -



- The research aimed to understand the challenges people face while searching for blood during emergencies and to identify opportunities for improving the blood donation experience through a digital platform .

Objectives –

-  Understand user behavior during blood emergencies .
 -  Identify the biggest challenges in finding blood donors .
 -  Explore how users currently search for blood .
 -  Measure users' willingness to register as blood donors .
 -  Discover the most valuable features users expect in a blood donation application .
- 

USER RESEARCH -



- To understand real-world challenges related to blood donation and emergency blood requests, a primary user survey was conducted using google forms. The survey aimed to identify user pain points, current behaviors , and expectations from a digital blood donation platform .

Research Method -

 **Method:** Online Google Form Survey

Participants: 22 Respondents

Target Audience:

Students

Working Professionals

General Public

Potential Blood Donors

Research Type: Quantitative Research

Duration: 2 Days

Research Outcome -

The survey revealed that users struggle to find verified blood donors quickly during emergencies and strongly prefer a platform that offers instant blood requests, nearby donor search, and trusted donor verification .

SURVEY INSIGHTS -

Key Findings from User Research -

The survey responses highlighted several pain points users face during blood emergencies. These insights helped shape the core features and user experience of **PulseLink**.

- **INSIGHT 1**

65% of participants have personally experienced, or know someone who has experienced, an emergency blood requirement .

- **INSIGHT 2**

75% of respondents identified **finding the correct blood group quickly** as the biggest challenge .

- **INSIGHT 3**

MOST USERS CURRENTLY DEPEND ON **FAMILY, FRIENDS, HOSPITALS, AND SOCIAL MEDIA** TO FIND BLOOD DONORS.

- **INSIGHT 4**

70% OF PARTICIPANTS SELECTED **EMERGENCY BLOOD REQUEST** AS THE MOST VALUABLE FEATURE.

COMPETITOR ANALYSIS -



Before designing PulseLink , existing blood donation platforms were analyzed to understand their strengths , limitations , and identify opportunities for improvement .

<u>Feature</u>	e-RaktKosh	Friends2Support	PulseLink
Find Nearby Donors	Limited	Yes	Yes
Emergency Blood Request	No	No	Yes
Verified Donor Profiles	Partial	No	Yes
Blood Bank Locator	Yes	No	Yes
Modern User Interface	Complex	Outdated	Clean & Simple
Quick Contact (Call/Message)	No	Limited	Yes

USER PERSONA -



Rahul Sharma

Age: 24 Years

Occupation: Software Engineer

Location: Delhi, India

Blood Group: A+

About -

Rahul is a working professional who believes in social causes and wants to donate blood whenever possible. However, during a family emergency, he struggled to find a verified blood donor quickly. He often relies on WhatsApp groups and hospital contacts, which takes valuable time.



Pain Points

- Difficulty finding the correct blood group .
- No trusted donor verification .
- Slow responses from social media groups .
- Limited information about nearby blood banks .



Needs

- Fast emergency request feature .
- Nearby verified donor search .
- Secure and trusted platform .
- Easy communication with donors .

" In an emergency, I don't want to waste time searching across different platforms. I need verified blood donors instantly . "

EMPATHY MAP -

THINKS (What does the user think ?)

“ Will I find the required blood in time ? ”

“ Can I trust this donor ? ”

“ I need a faster way to search . ”

“ Every minute matters during an emergency . ”



DOES (What does the user do?)

Calls family and friends .

Posts requests on WhatsApp/social media .

Contacts hospitals and blood banks .

Searches online for blood donors .



SAYS (What does the user say ?)

“ I need O+ blood urgently . ”

“ Is there anyone nearby who can help ? ”

“ Hospitals are asking me to arrange donors . ”

“ I wish there was one app for this . ”

FEELS (What does the user feel ?)

Stressed during emergencies .

Anxious about delays .

Hopeful when verified donors are available .

Relieved after finding the right donor .

USER JOURNEY MAP -



This journey illustrates the experience of a user trying to find blood during an emergency and highlights the opportunities where PulseLink improves the process .

Stage	User Action	Thoughts	Pain Points	PulseLink Solution
Awareness	Learns that blood is urgently needed	" I need blood immediately . "	Panic and confusion	Quick access to Emergency Request
Search	Contacts family, friends & hospitals	" Where can I find the right donor ? "	Time-consuming process	Find Nearby Verified Donors
Decision	Chooses a donor	" Can I trust this donor ? "	Lack of verification	Verified Donor Profiles
Action	Contacts the donor	" I hope they respond quickly . "	Slow communication	One-tap Call & Message
Success	Blood is arranged	" The process was much easier . "	Relief	Fast and reliable experience

IDEATION -



After defining the problem, multiple ideas were explored to identify the most effective solution for users during blood emergencies. Different concepts were brainstormed before selecting the final direction for PulseLink .

Ideas -

-  Live blood donor search
-  One-tap emergency blood request
-  Blood bank locator
-  Verified donor profiles
-  Instant call & chat
-  Emergency notifications
-  Donation history & eligibility

INFORMATION ARCHITECTURE -

The information architecture was designed to ensure users can quickly access essential features during emergencies. The navigation is simple, intuitive, and focused on minimizing the number of steps required to find blood donors .

-  **Pulselink**



-  Home

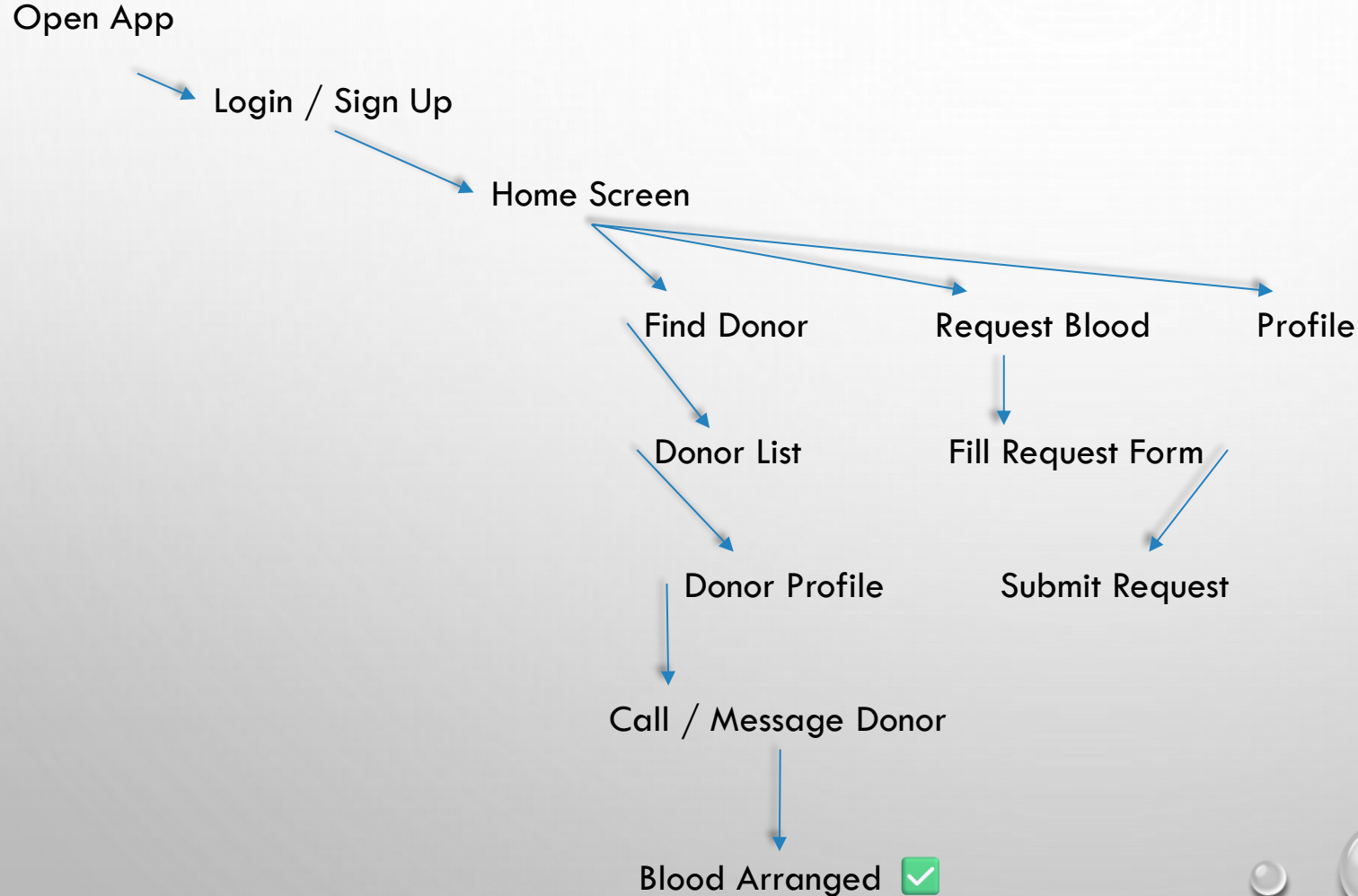


-  Find donor
-  Request blood
-  Blood banks
-  Profile

Reduce the number of steps required to access life-saving features .

USER FLOW -

The user flow illustrates the step-by-step journey a user follows to complete the primary task of finding blood donors or creating an emergency blood request using PulseLink .



WireFrames

1. Login screen

 Pulselink

Welcome back

Email

Password

Login

Sign up

2. Home Screen

Welcome 🖐️

 Search

Request Blood

- Find Donor
- Blood Bank

- Nearby
 - Profile
- Recent Requests

3. Find Donor

Find Donor

O+ A+ B+

Akash
3 Km
Call Msg

4. Blood Request

Request Blood

Patient Name

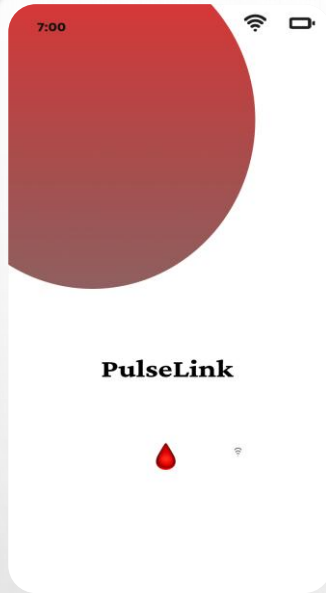
Blood Group

Hospital

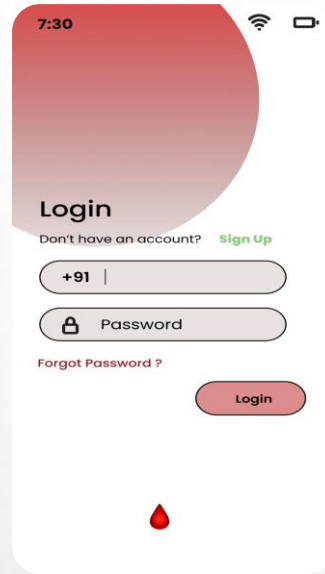
Contact

Submit

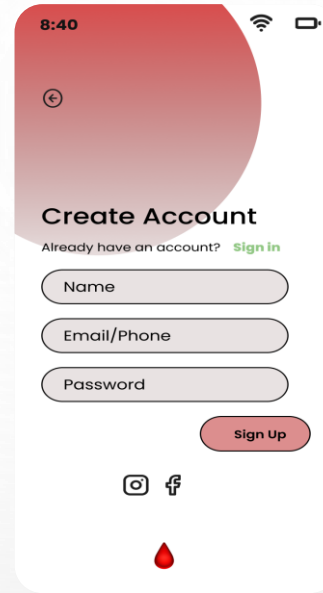
USER INTERFACE -



Splash Screen



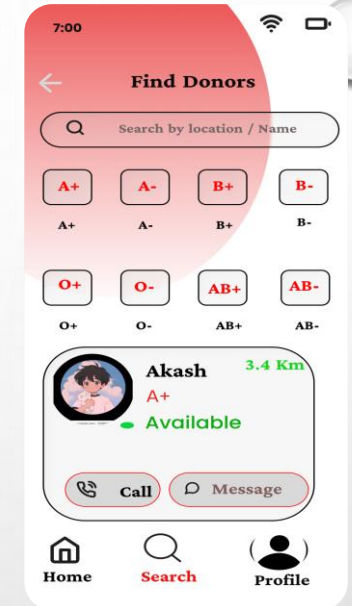
Login Page



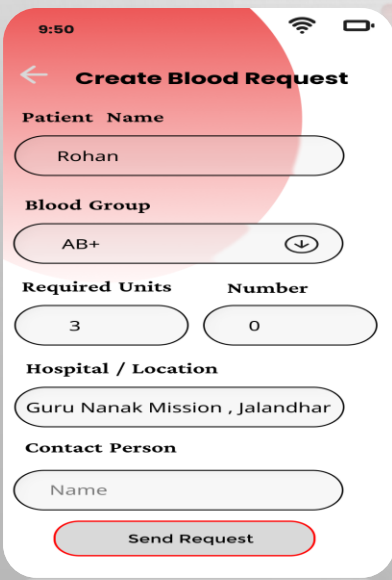
Sign Up Page



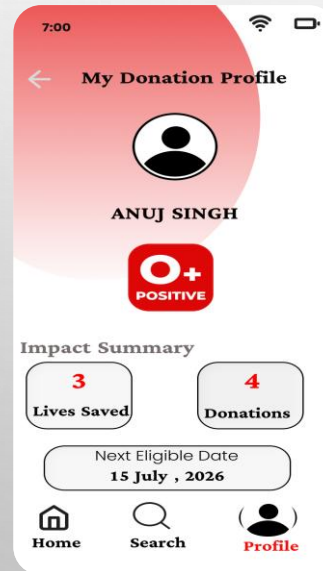
Home



Find Donor



Blood Request

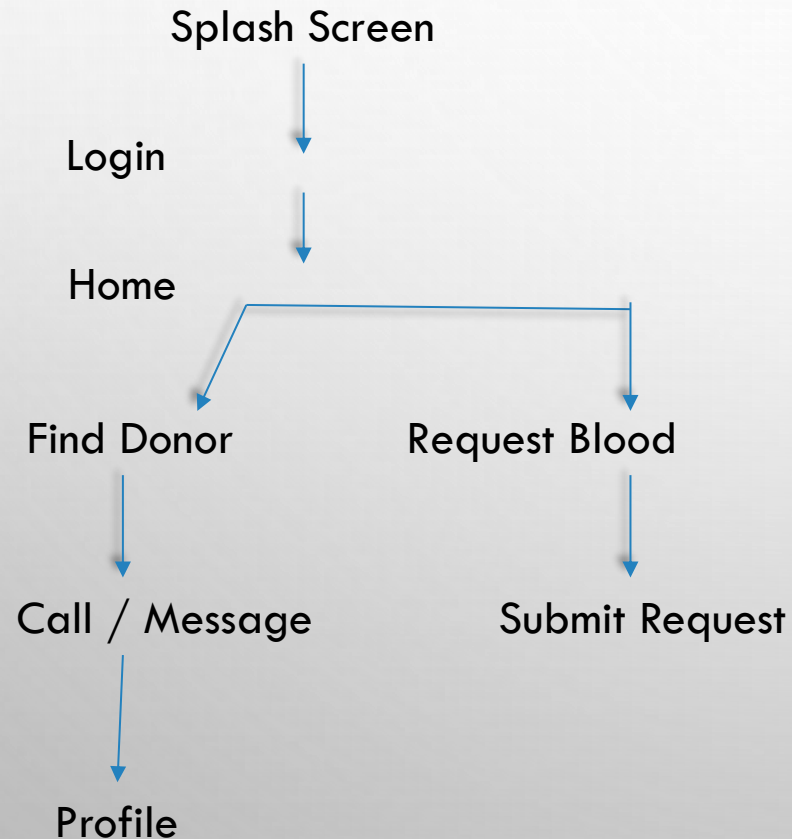


Profile

INTERACTIVE PROTOTYPE -

After completing the User interface, an interactive prototype was created in Figma to simulate the real user experience. This allowed the navigation flow and interactions to be tested before development .

Prototype Flow -



USABILITY TESTING



Testing details

Participants : 5 users (*selected from survey respondents*)

Method : prototype testing (Figma)

Tasks given :

- Login to the application .
- Find a nearby blood donor .
- Submit an emergency blood request .
- Open the profile page .



Improvements Suggested –

- Add real-time notifications for request updates .
- Include live blood availability status .
- Add dark mode for better accessibility .
- Enable in-app chat between donors and recipients .
- Add multi-language support .

User Feedback



Positive Feedback

- Easy and intuitive navigation .
- Emergency Request button is easy to find .
- Clean and modern interface .
- Blood group filters are simple to use .
- Consistent colour scheme improves readability .

CONCLUSION -

- PulseLink was designed using a **human - centred design process** to address one of the most critical healthcare challenges — finding blood donors during emergencies. Through user research, survey analysis , competitor analysis , and iterative design, the solution focuses on reducing search time, improving trust through verified donor profiles , and providing a simple , accessible user experience .
- The final prototype demonstrates how a research-driven approach can transform a real - world problem into an intuitive and meaningful digital solution .

Project Outcomes ---

1. Research-driven design decisions
2. Clean and consistent user interface
3. Emergency-first navigation
4. Verified donor discovery
5. Faster blood request process



**THANK
YOU!**